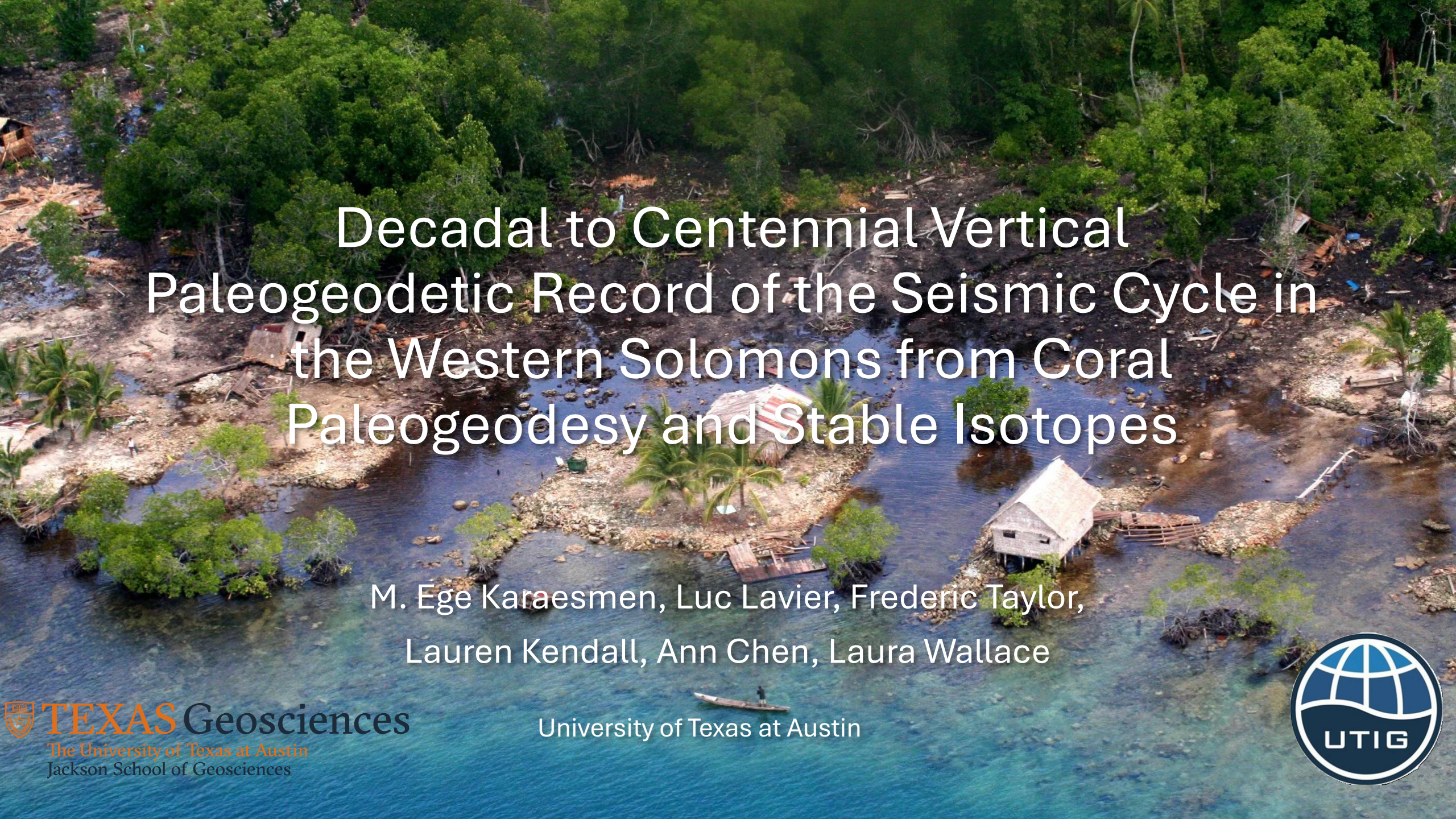
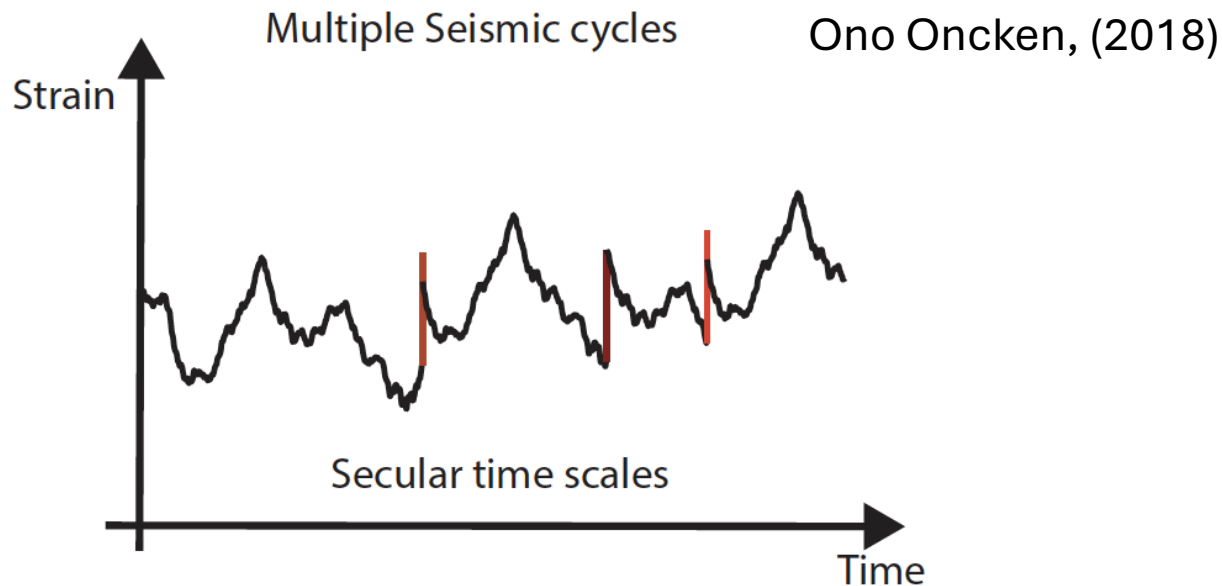
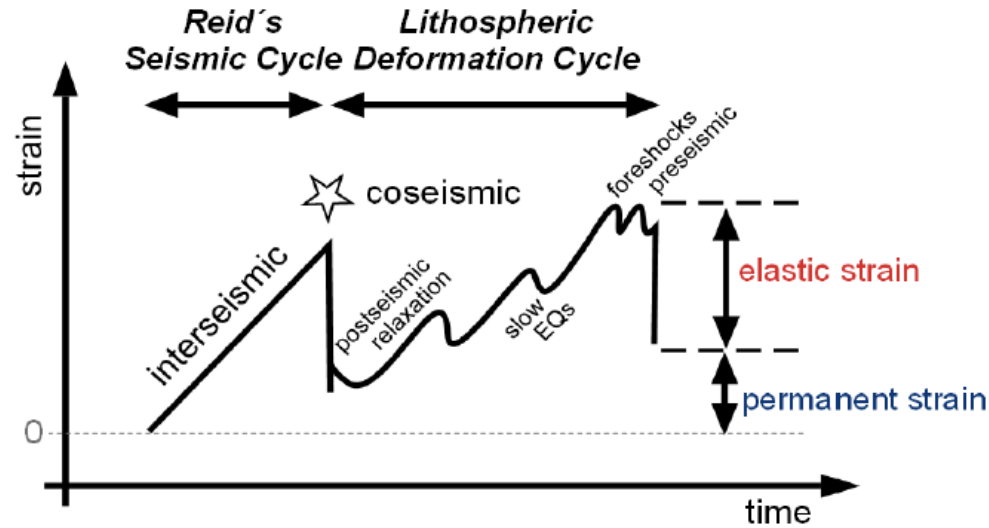




Decadal to Centennial Vertical Paleogeodetic Record of the Seismic Cycle in the Western Solomons from Coral Paleogeodesy and Stable Isotopes

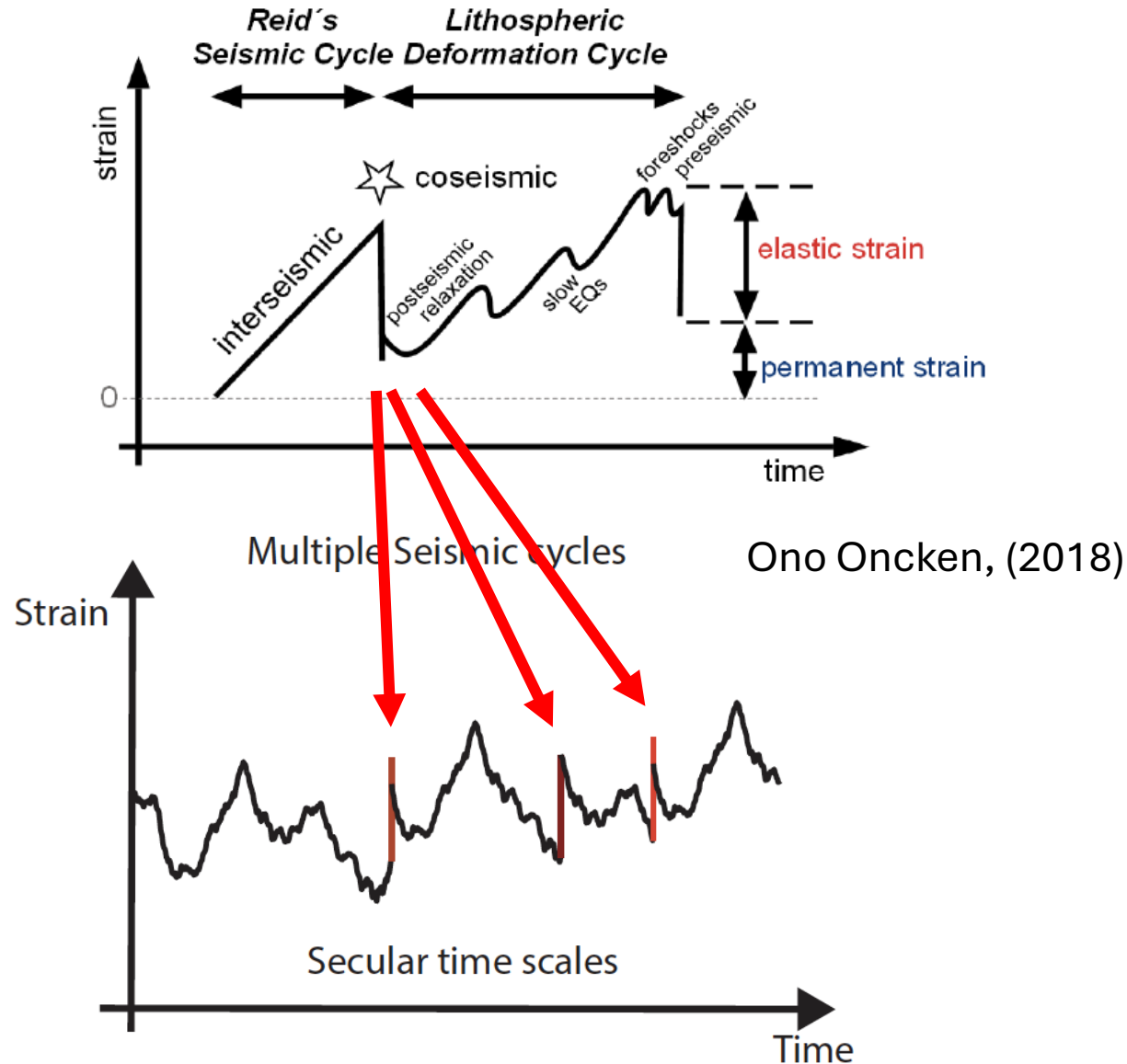
M. Ege Karaesmen, Luc Lavier, Frederic Taylor,
Lauren Kendall, Ann Chen, Laura Wallace

Earthquake Cycles Reconstruction



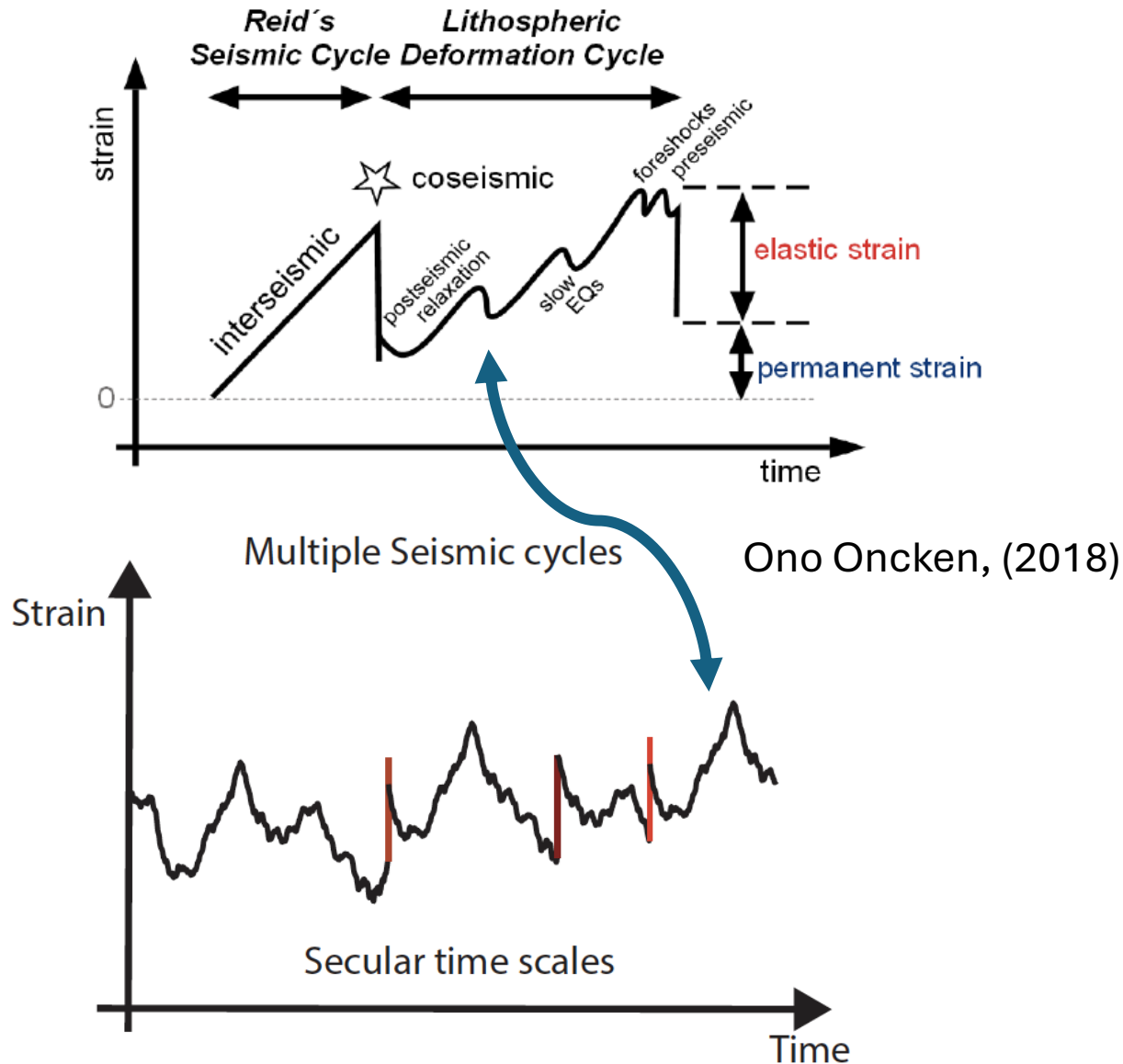
- Reconstruct vertical tectonics over 100s to 1000s years using coral paleogeodesy
- Including tectonic motion over multiple seismic cycles
- Infer viscoelastic properties
- Slow slip events + Precursory

Earthquake Cycles Reconstruction

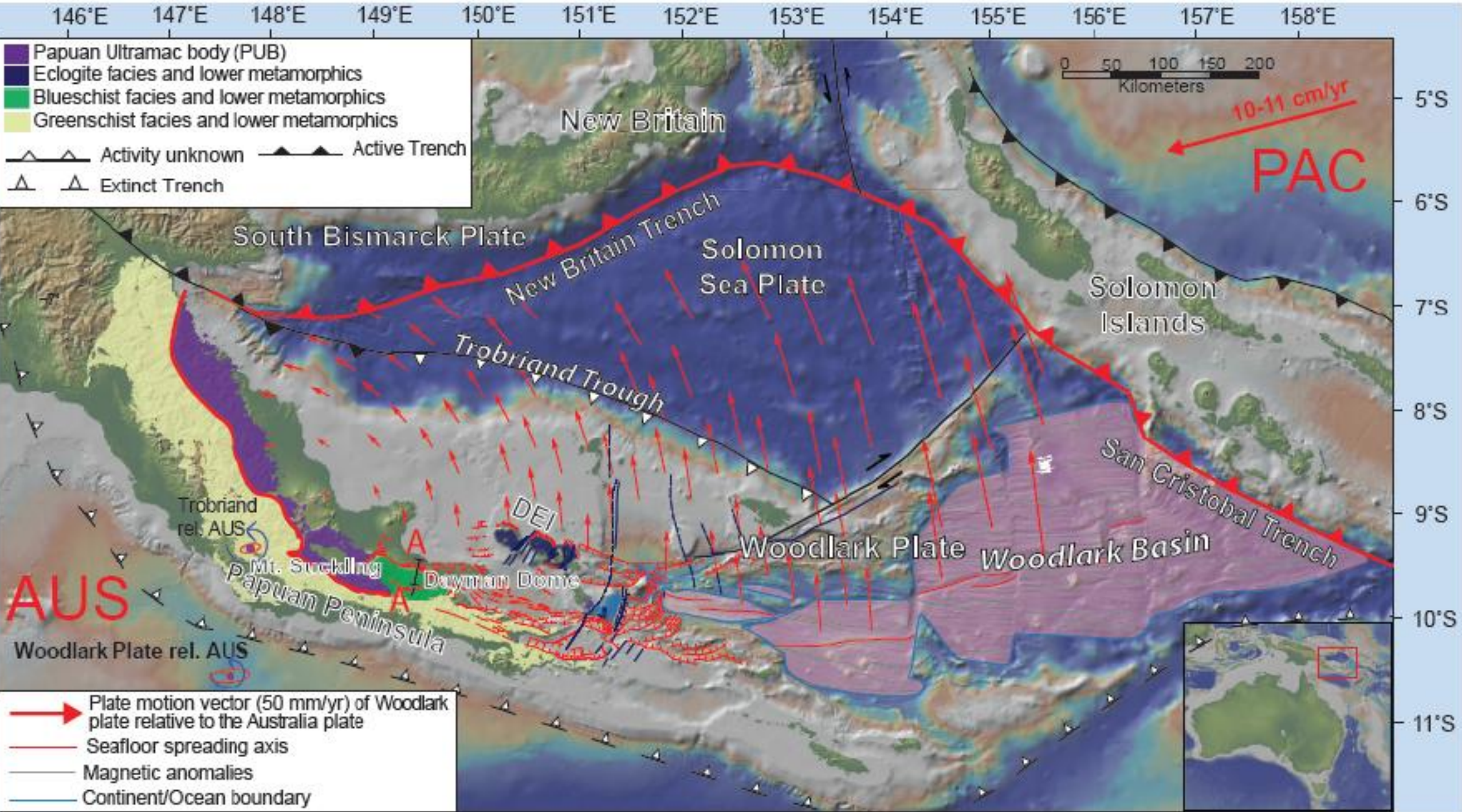


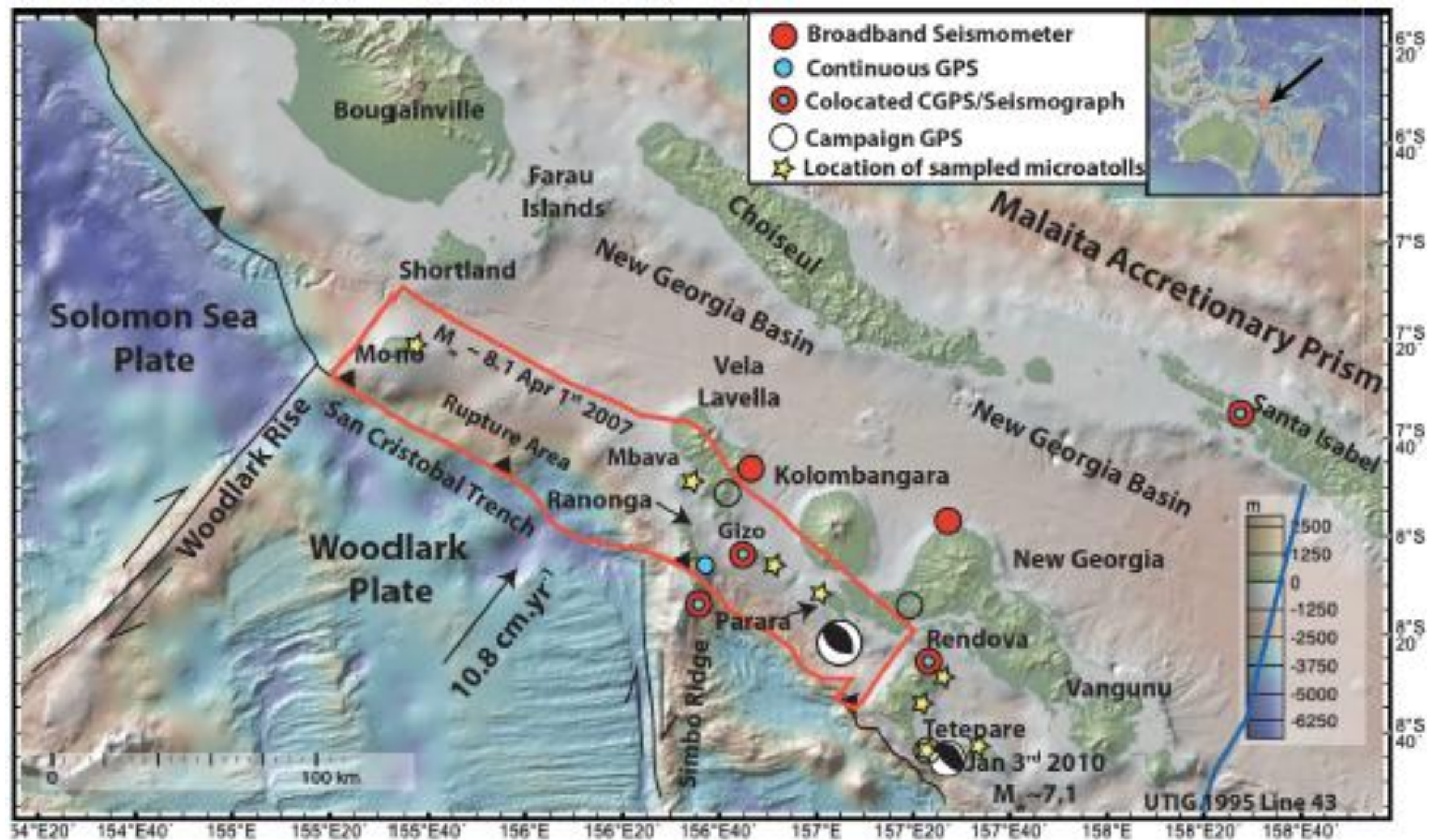
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Earthquake Cycles Reconstruction



- Reconstruct vertical tectonics over 100s to 1000s years using coral paleogeodesy
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After Mw 8.1 2007



Luc Lavier measuring the vertical position of the sampled coral, near Ranongga Island in 2012.



UT Austin doctoral student Kaustubh Thirumalai inspecting corals near the Solomon Islands' Ranongga Island in 2015.

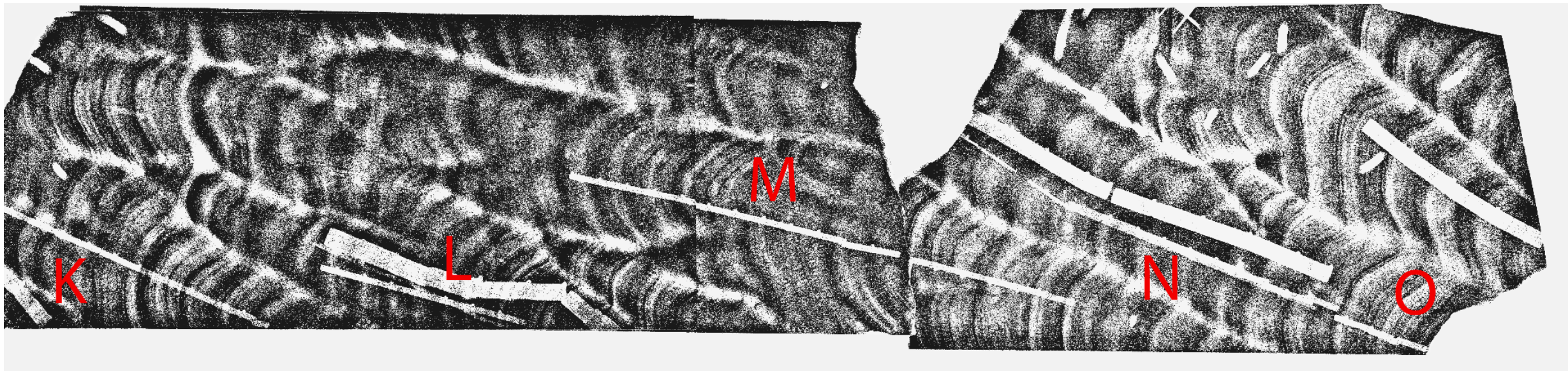
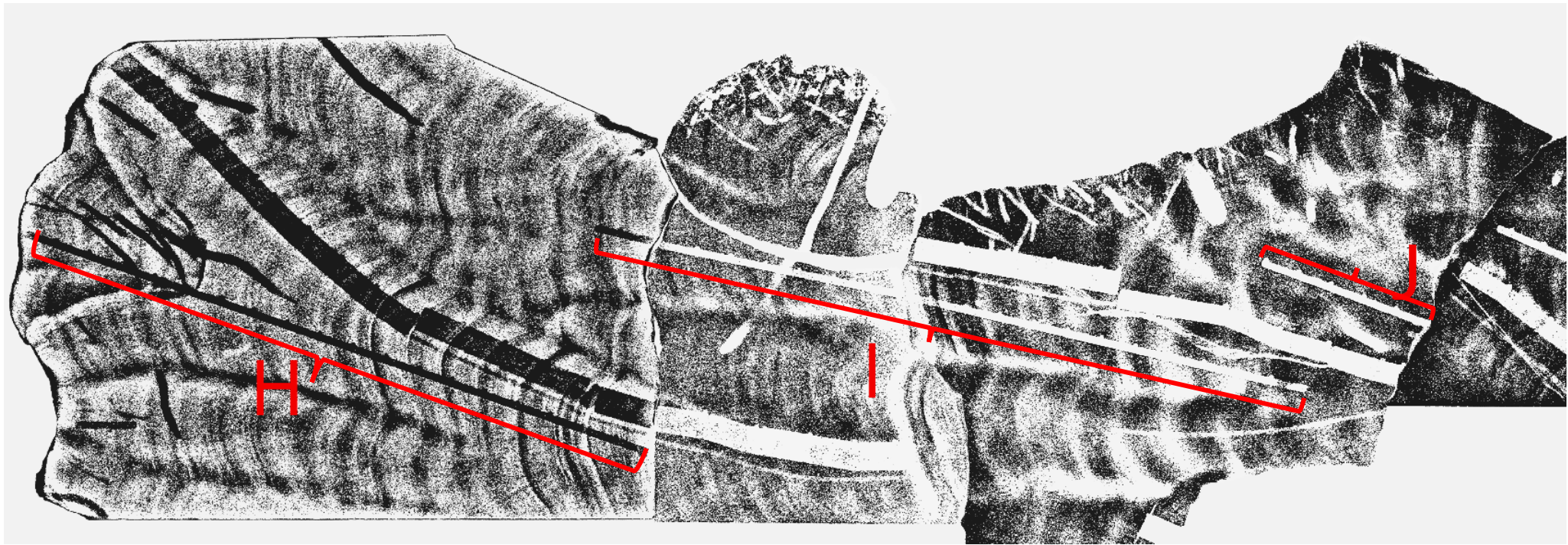
After Mw 8.1 2007



Fred Taylor (UT Austin) collects samples from corals in the Solomon Islands, after 2007 Earthquake uplift. (2015)

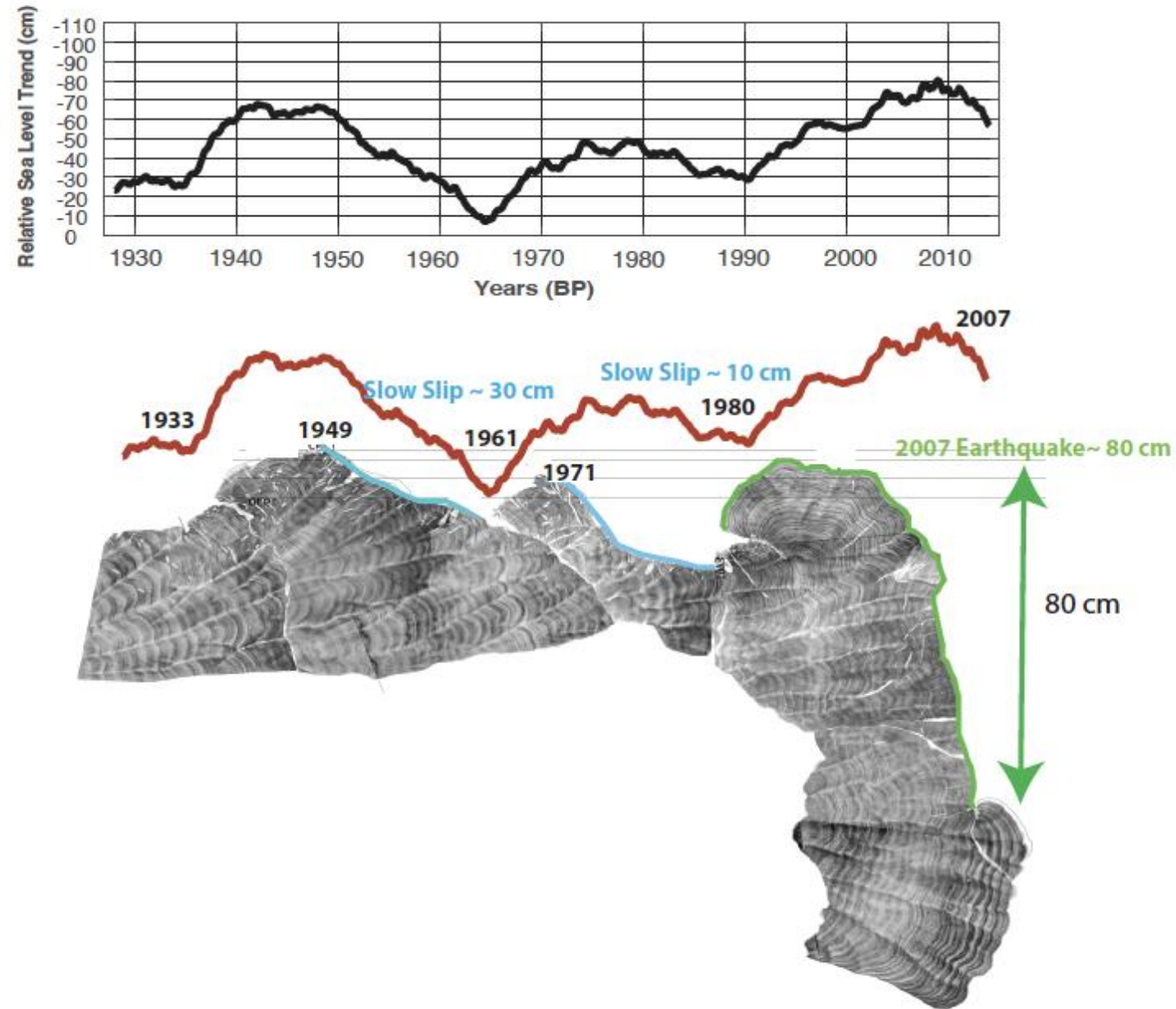


The sample taken in 2012 from Parara Island and analyzed in 2024 at Austin.

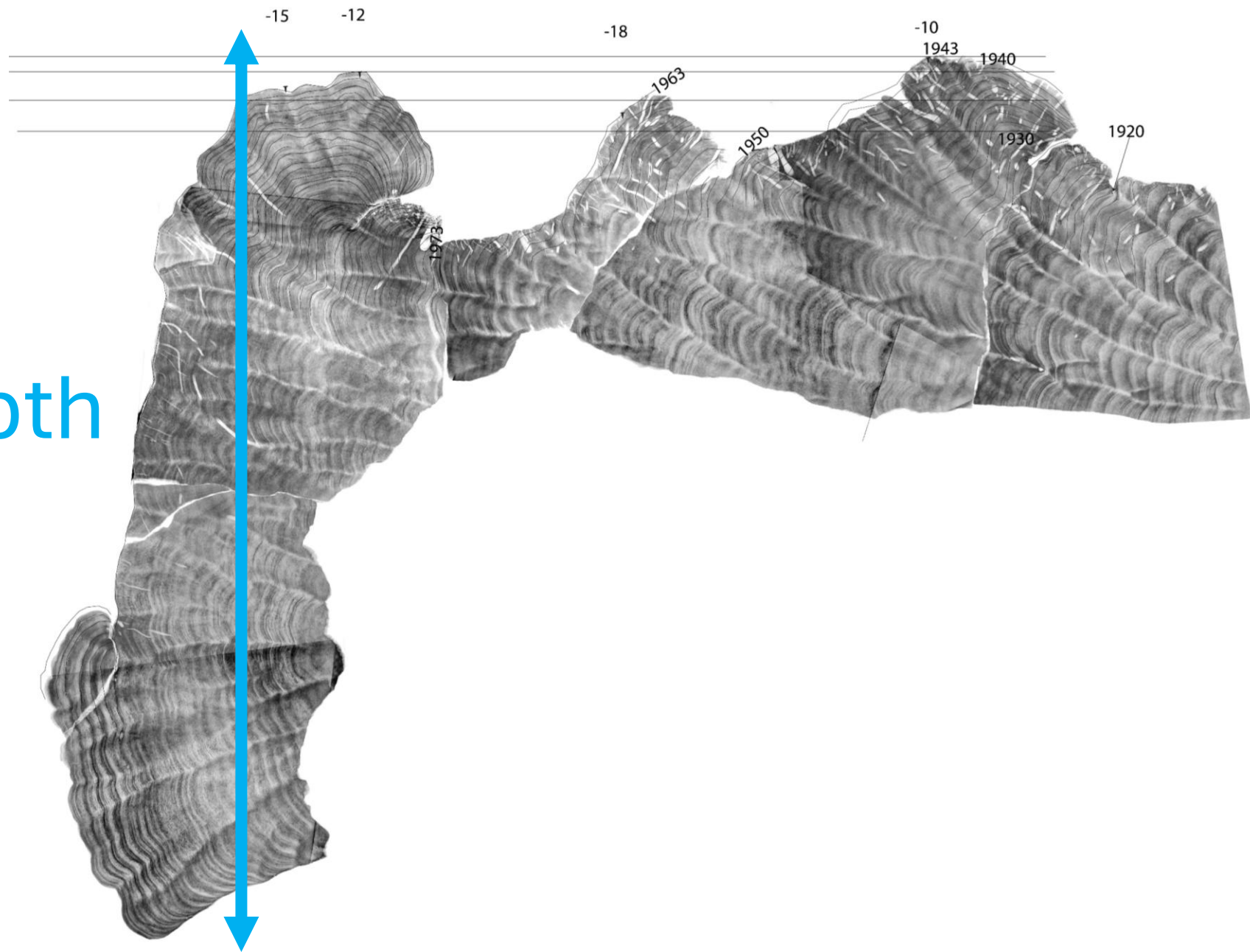


MORPHOLOGY

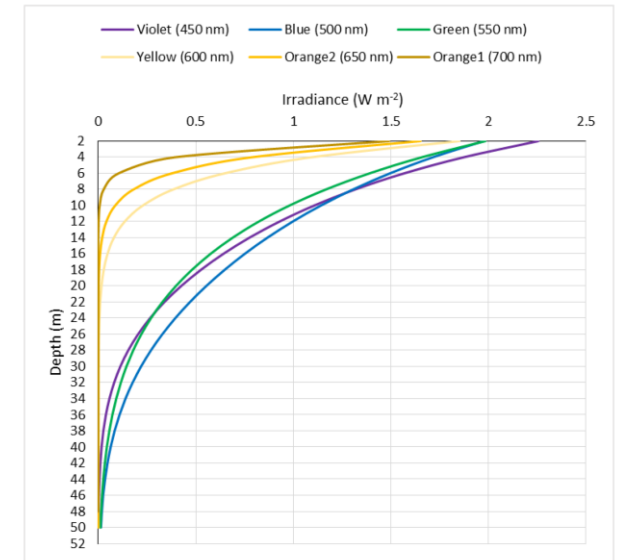
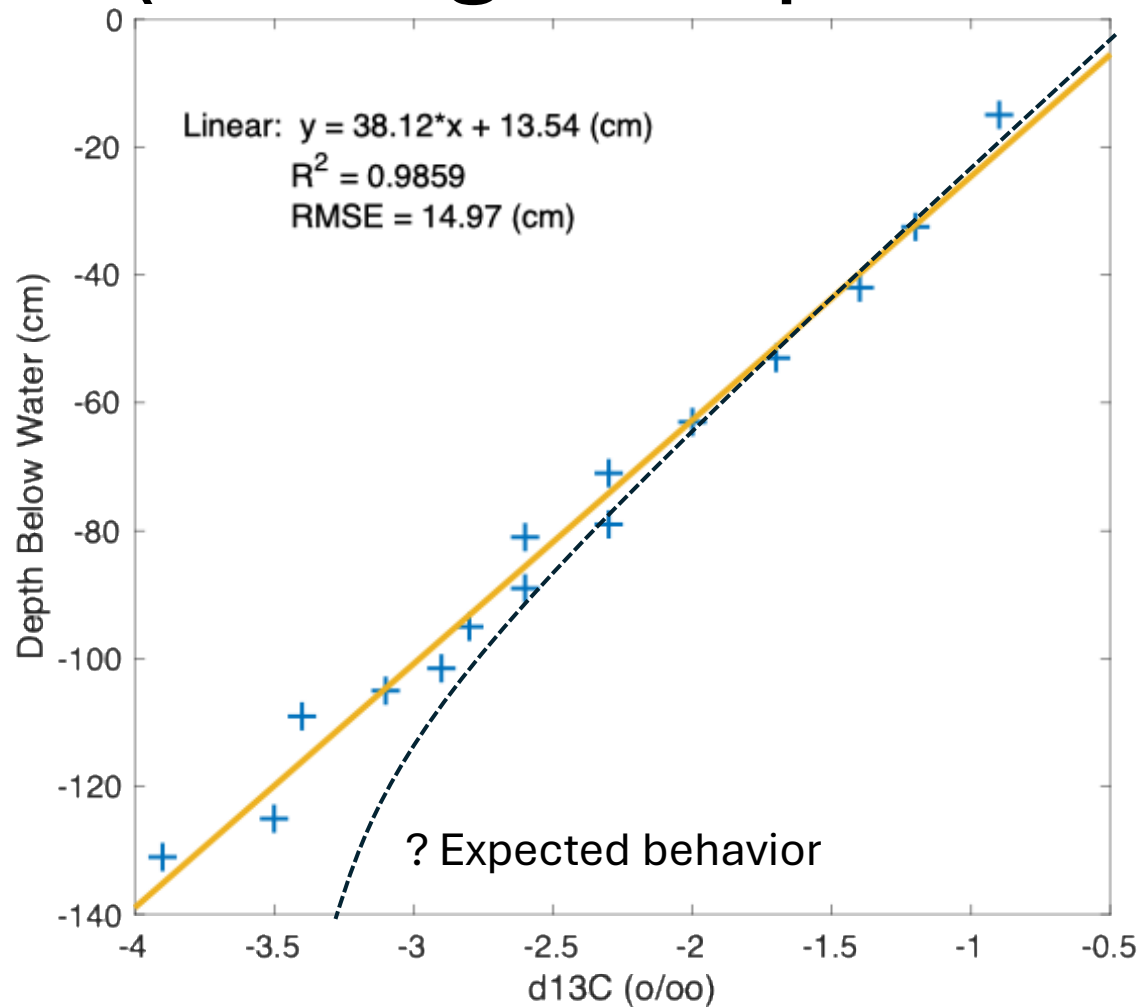
Sea Level Change Relative to the Coral Surface



Depth



Confirmed with additional data (Blue light/ euphotic zone)

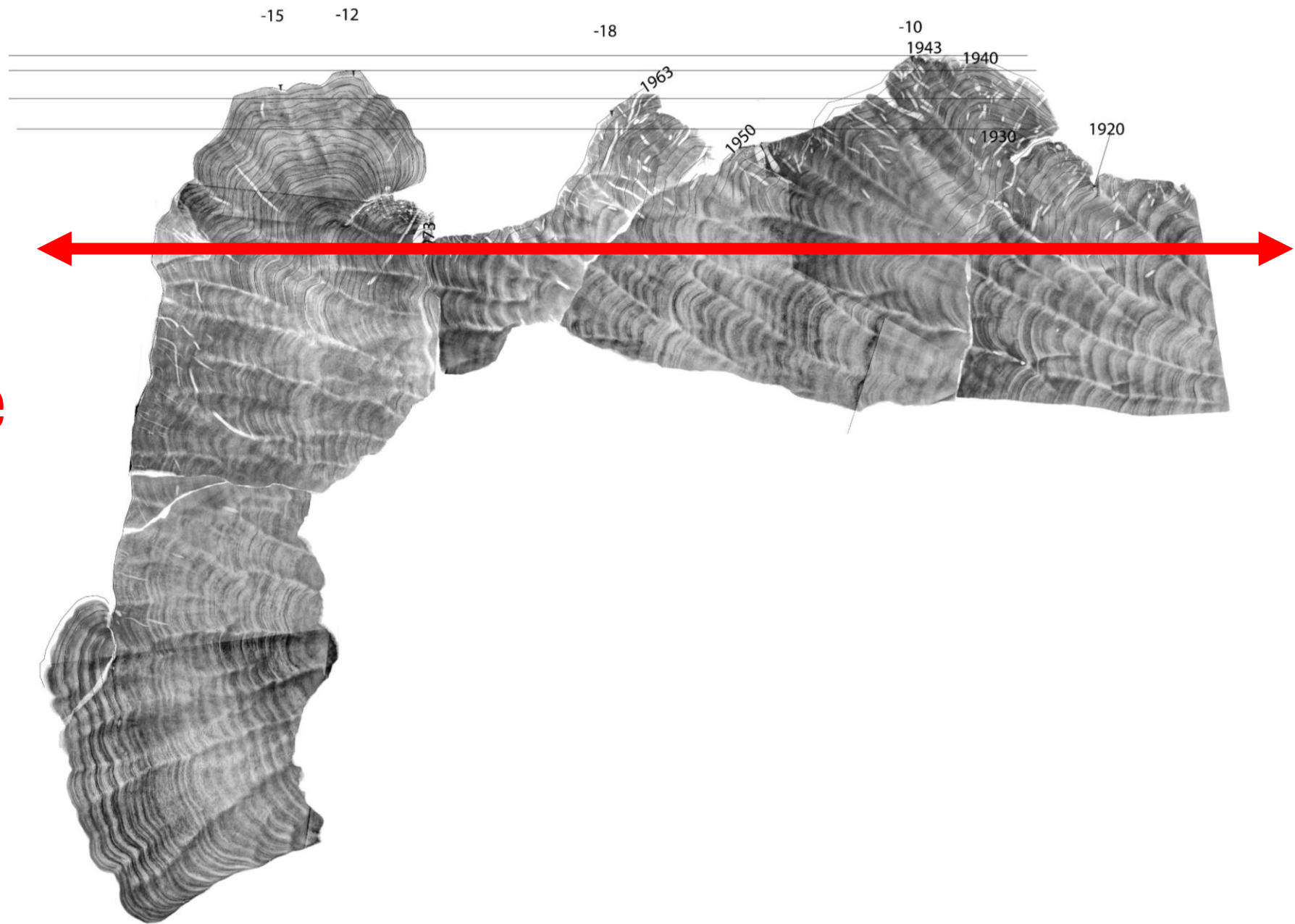


Abdelrahman, Mohamed A. 2016

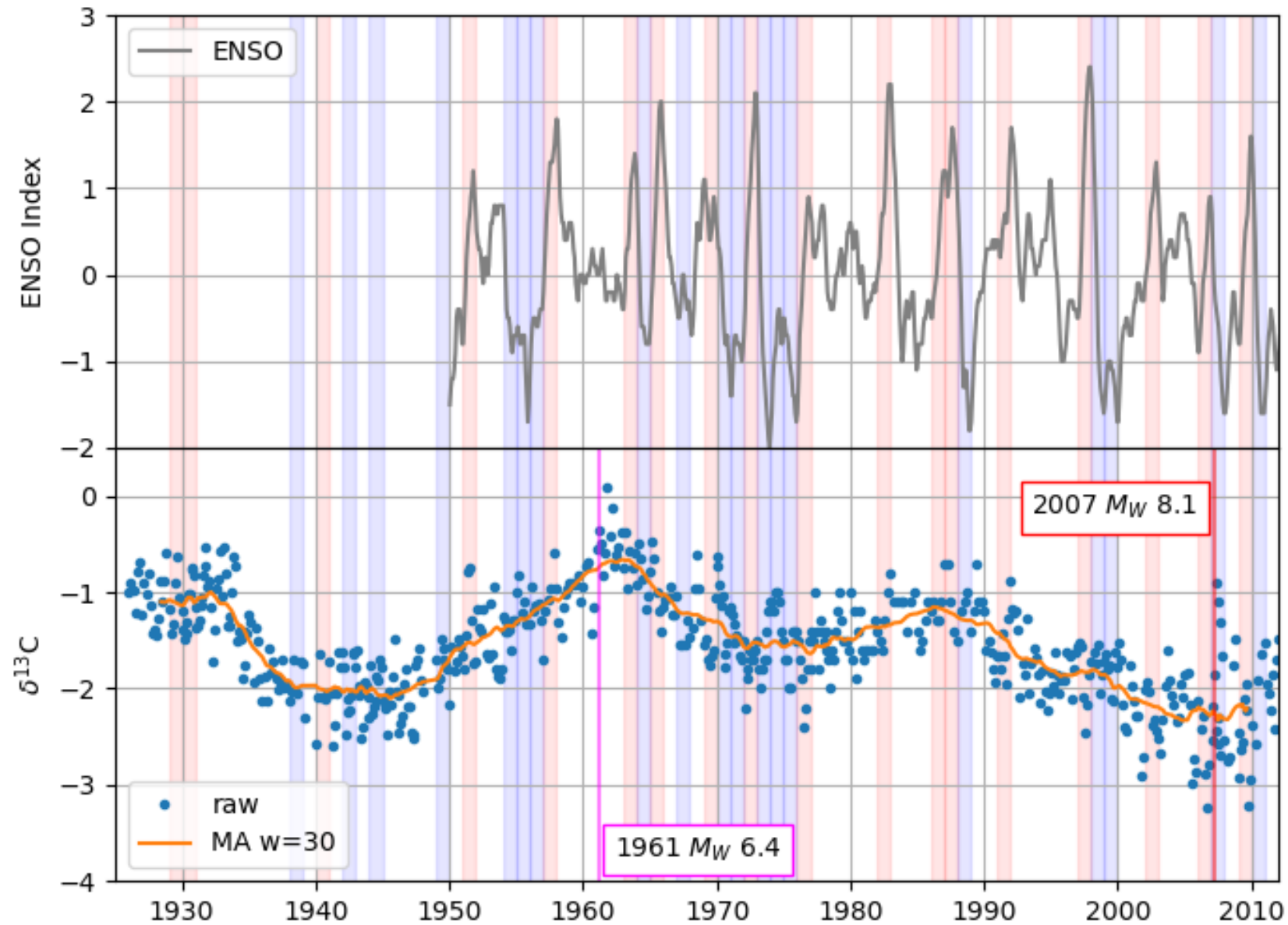
HYPOTHESIS

- Plankton rejects heavier carbon ^{13}C when sunlight is less intense (up to 50 m depth for blue light).
- The excess ^{13}C is the used by the coral to build its skeleton.
- The relationship looks linear in shallow water but must be exponential and saturate at some depth.

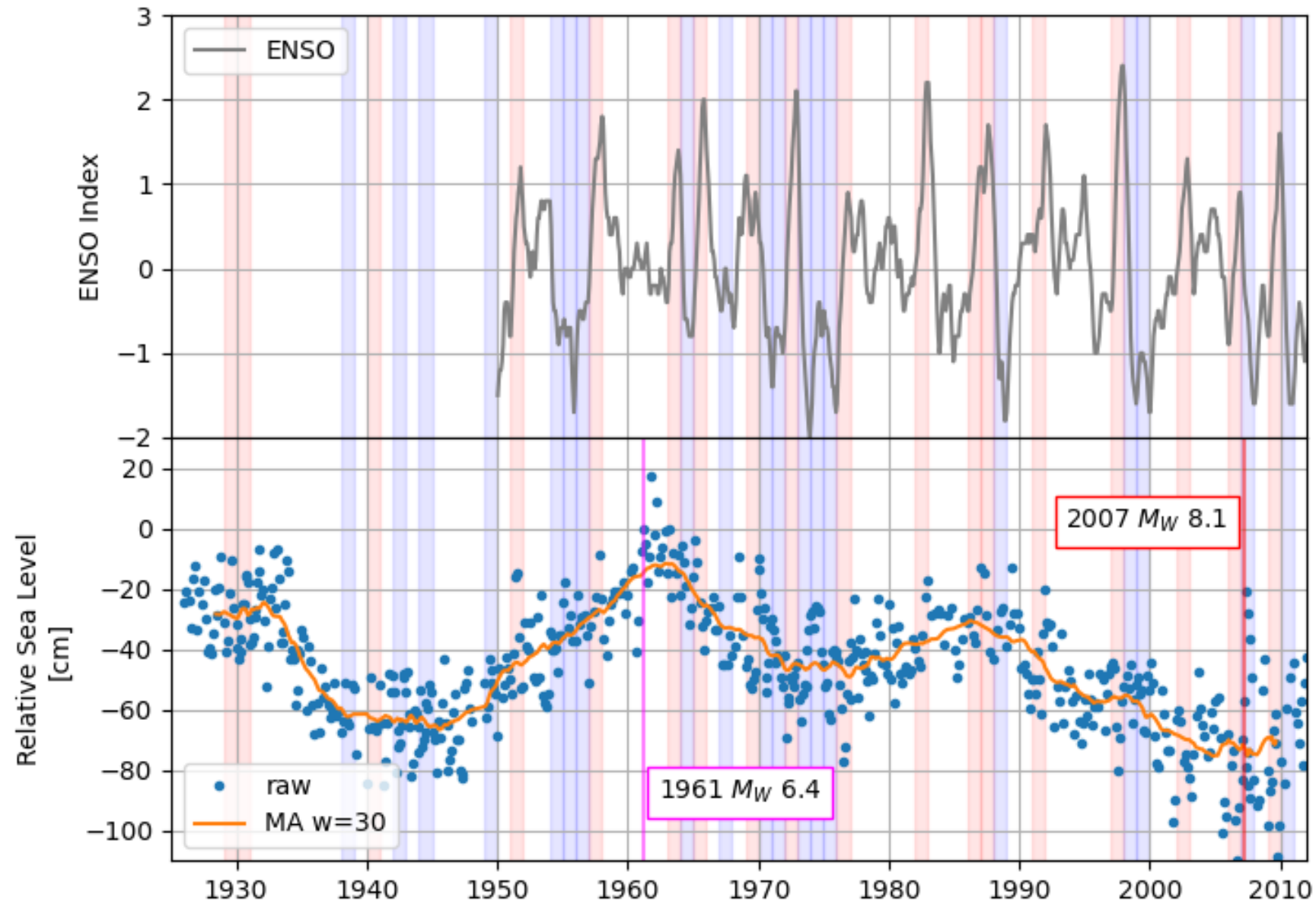
Age



$\delta^{13}\text{C}$ in Coral



Relative Sea Level Trend



Relative Sea Level Trend

